

RESEARCH INTEREST

Machine Learning

- *Data-Centric AI; Reinforcement Learning; Extrapolation; Robustness; Causality; AI for Social Good*

TECHNICAL SKILLS

Skills

- *Python; PyTorch; Hugging Face Transformers; SQL; Git; Linux; data pipelines; LLM synthetic data; benchmarking/evaluation; NLP*

EDUCATION

Swiss Federal Institute of Technology (EPFL)

Lausanne, Switzerland

- *M.Sc. in Computer Science (Specialization in Computer Science Theory)*

Sept. 2024 – Present

- **Coursework:** Machine Learning; Advanced Algorithms; Causal Thinking; Mathematics of Data: From Theory to Computation; Markov Chains and Algorithmic Applications; Deep Learning; Reinforcement Learning; Modern Natural Language Processing; Distributed Algorithms; Topics in Theoretical Computer Science
- Completed mandatory military service (Mar. 2025 – Jun. 2025)

National Taiwan University

Taipei, Taiwan

- *M.Sc. in Data Science*

Feb. 2023 – June 2024

- Paused studies after June 2024 to pursue a master's degree at EPFL
- **Coursework:** Machine Learning; Natural Language Processing; Statistical Foundations of Data Science (I); Computational Methods for Data Science; Big Data Systems

National Taiwan University of Science and Technology

Taipei, Taiwan

- *B.Sc. in Applied Science and Technology, Major in Computer Science*

Sept. 2018 – June 2022

- Graduated 1st in the program (2022)
- **Coursework:** Statistics (I); Probability and Statistics; Discrete Mathematics; Linear Algebra; Calculus (I); Algorithms; Data Structures; Database Systems; Operating Systems; Digital Logic Design; Computer Networks; Webpage Development; Object-Oriented Programming; Introduction to Computers; Value of AI and Data; Introduction to Fuzzy Systems; Machine Learning Foundations

RESEARCH AND INDUSTRY EXPERIENCE

MLO Lab, EPFL

Lausanne, Switzerland

- *Semester Project Student*

Sept. 2025 – Present

- **Advisors:** Martin Jaggi; El Mahdi Chayti
- Studying looped/recurrent Transformers for length extrapolation, focusing on learned stochastic stopping, OOD robustness, and stability across inference-time computation depths.

Robust Machine Learning Group, Max Planck Institute for Intelligent Systems

Tübingen, Germany

- *Research Intern*

June 2025 – Aug. 2025

- **Advisors:** Wieland Brendel; Patrik Reizinger
- Designed controlled experiments on OOD extrapolation in looped Transformers, analyzing the effects of initialization, data order, and loop budget on generalization stability.

CKIP Lab, Academia Sinica & IR Lab, National Taiwan University

Taipei, Taiwan

- *Research Assistant*

Feb. 2023 – June 2024

- **Advisors:** Wei-Yun Ma; Pu-Jen Cheng
- Developed data-weighting methods for LLM-generated synthetic text-classification data, improving robustness under source-target distribution shift.

Industrial Technology Research Institute

Taipei, Taiwan

- *Intern*

July 2021 – Sept. 2021

- Engineered short- and long-term load forecasting methods for Taiwan Power Company.

Wireless System Lab, National Taiwan University of Science and Technology

Taipei, Taiwan

- *Research Assistant*

Sept. 2019 – Jan. 2021

- **Advisor:** Chin-Ya Huang
- Devised an MTC scheme for dynamic CBRS channel allocation.
- Implemented RLNC and Galois field arithmetic in P4; simulated with ONOS/Mininet.

PAPERS

- [P1] **Hsun-Yu Kuo**, El Mahdi Chayti, Patrik Reizinger, Wieland Brendel & Martin Jaggi, “Stabilizing Extrapolation in Looped Transformers via Learned Stochastic Stopping”. In *ICML 2026 Workshop on Resource-Adaptive Foundation Model Inference* (AdaptFM 2026) [📄](#)
- [P2] **Hsun-Yu Kuo**, Yin-Hsiang Liao, Yu-Chieh Chao, Wei-Yun Ma & Pu-Jen Cheng, “Not All LLM-Generated Data Are Equal: Rethinking Data Weighting in Text Classification”. In *Proceedings of the 13th International Conference on Learning Representations* (ICLR 2025) (**Spotlight**, top 5.1%) [📄](#)
- [P3] **Hsun-Yu Kuo**, Szu-Yu Liu, Chin-Ya Huang, Yu-Chi Chen & Meng-Hua Xie, “Reliable Data Transmission through Private CBRS Networks”. In *Preprint* (2023) [📄](#)
- [P4] **Hsun-Yu Kuo**, T.-W. Liu, Y.-P. Huang, *et al.*, “Differential Diagnostic Value of Machine Learning–Based Models for Embolic Stroke”. *Clinical and Applied Thrombosis/Hemostasis* (2023)
- [P5] Hsuan-Min Wang, Yo-Ping Huang, **Hsun-Yu Kuo**, *et al.*, “The modified spatial context memory test for assessing cognitive aging in middle-aged and older adults”. *Neuroscience Research* (2019)

TEACHING EXPERIENCE

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|---|-------------------------------|
| Teaching Assistant, National Taiwan University | Taipei, Taiwan |
| ▪ <i>System Programming</i> | <i>Sept. 2023 – Jan. 2024</i> |
| ◦ Led TA sessions and supported 100+ students. | |
| ◦ Designed a context-switch simulation using non-local jumps and signals for a course assignment. | |

HONORS AND AWARDS

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| Phi Tau Phi Scholastic Honor Society of the Republic of China | Taiwan |
| ▪ <i>Honorary Membership</i> | <i>2022</i> |
| Mengjia Longshan Temple Scholarship | Taiwan |
| ▪ <i>Academic Excellence Scholarship (Scholarship for Excellent Academic Performance)</i> | <i>2021</i> |
| Academic Excellence Award | Taiwan |
| ▪ <i>Seven consecutive awards for excellent academic performance</i> | <i>2018–2022</i> |
| Kanagawa International Science Forum | Kanagawa, Japan |
| ▪ <i>1st Place (Outstanding Poster Presentation Award) – Engineering Category</i> | <i>2017</i> |
| National Skills Competition | Taiwan |
| ▪ <i>1st Place (Golden Award) – Software Applications Development</i> | <i>2016</i> |

PROFESSIONAL SERVICE

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| Conference Reviewer |
| ▪ <i>ICLR 2026, NeurIPS 2026</i> |
| Volunteer |
| ▪ <i>SIGIR 2023</i> |